

Evaluating NLDR Layouts using a Model in the High-dimensional Data Space

Jayani P. Gamage

Econometrics & Business Statistics, Monash University
and

Dianne Cook

Econometrics & Business Statistics, Monash University
and

Paul Harrison

MGBP, BDInstitute, Monash University
and

Michael Lydeamore

Econometrics & Business Statistics, Monash University
and

Thiyanga S. Talagala

Statistics, University of Sri Jayewardenepura

September 21, 2026

Abstract

Nonlinear dimension reduction (NLDR) techniques such as t-SNE and UMAP provide low-dimensional representations of high-dimensional data, capturing complex structures in a concise visual summary. While NLDR has an important role in data analysis, these methods can badly miss the mark by exaggerating structures or suggesting spurious patterns. Different methods and hyper-parameter choices can produce wildly different representations, making it difficult to decide which layouts are reasonable representations and which are misleading. To help, we have developed a procedure that treats the low-dimensional NLDR layout as a model that can be lifted up into the original high-dimensional space. This enables us to assess goodness of fit by computing residuals and visualizing the fitted model overlaid on the data using grand tours. This approach allows us to quantitatively compare multiple layouts, determine which representations are reasonable, and assess whether an embedding captures the data structure throughout the space, only in certain subspaces, or performs poorly overall.

Keywords: high-dimensional data, dimension reduction, data visualization, statistical graphics, tour, unsupervised learning

1 Introduction

Nonlinear dimension reduction (NLDR) is popular for making a convenient low-dimensional (k -D) representation of high-dimensional (p -D) data ($k < p$). Recently developed methods include t-distributed stochastic neighbor embedding (tSNE) (Maaten & Hinton 2008), uniform manifold approximation and projection (UMAP) (McInnes et al. 2018), potential of heat-diffusion for affinity-based trajectory embedding (PHATE) algorithm (Moon et al. 2019), large-scale dimensionality reduction using triplets (TriMAP) (Amid & Warmuth 2022), and pairwise controlled manifold approximation (PaCMAP) (Wang et al. 2021).

However, the representation generated can vary dramatically from method to method, choice of hyper-parameter, or even random seed, as illustrated by Figure 1. The choice may result in different procedures being used in the downstream analysis or different inferential conclusions. Various academics have expressed concerns about current practices and procedures for choosing (e.g., Irizarry (2024), Chari & Pachter (2023)). The research described here provides new numerical and visual tools to help choose a suitable layout.

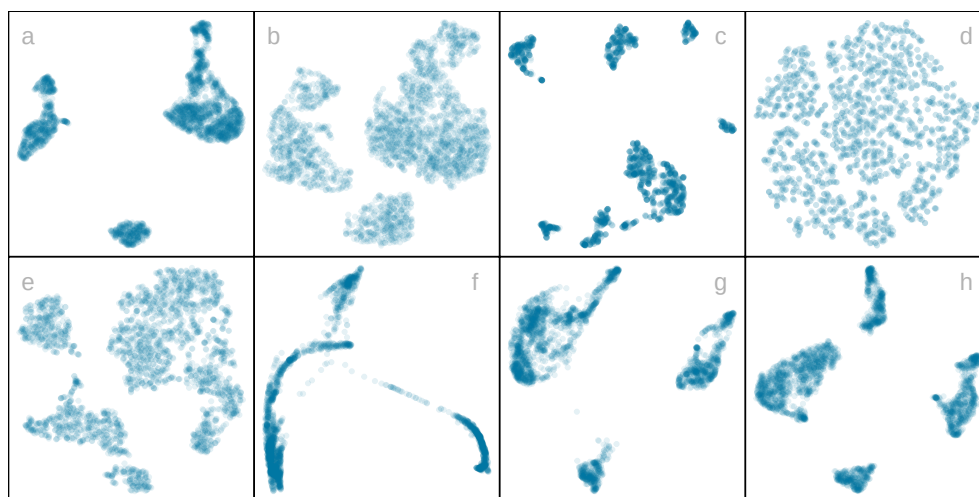


Figure 1: Eight different NLDR representations of the same data, produced by tSNE, UMAP, PHATE, TriMAP and PaCMAP with a variety of hyper-parameter choices. The variety in layouts makes it difficult to choose which best represents the data distribution.

The paper is organized as follows. Section 2 gives more details about NLDR and high-dimensional data visualization. Sections 3, 4, 5 describe the new methodology, illustrated using a simulated data example. Section 6 explains how to use this approach to choose from a selection of layouts. Section 7 extends the approach to predict the layout positions of new observations. Two applications are in Section 8. Section 9 discusses limitations and future directions. In addition, there is an R package `quollr` which implements the methods.

2 Background literature on dimension reduction

Making useful low-dimensional (k - D) layouts of high-dimensional (p - D) data has historically been achieved using multidimensional scaling (MDS) (Kruskal 1964). In MDS, the k - D layout is constructed by minimizing a stress function that compares distances between points in p - D with potential distances between points in k - D . Various formulations of the stress function result in non-metric scaling (Saeed et al. 2018) and isomap (Silva & Tenenbaum 2002). Principal components analysis (PCA) produces the same layout as classical MDS if interpoint distances are Euclidean. Recent overviews of MDS, PCA and statistically conducting linear and nonlinear dimension reduction can be found in Borg & Groenen (2005), Johnstone & Titterton (2009), Jolliffe (2011), Koch (2013).

The relatively recent NLDR methods, tSNE, UMAP, PHATE, TriMAP, and PaCMAP, can be considered new approaches tackling the same problem as MDS. The methods tSNE, UMAP, TriMAP, and PaCMAP are similar in that they determine the k - D representation by minimizing the divergence between two inter-point distance distributions. PHATE is different and is based on a diffusion process spreading to capture geometric shapes that

include both global and local structure. (See [Coifman et al. \(2005\)](#) for more details on diffusion processes.)

The array of layouts in Figure 1 illustrate what can result from choosing different methods, tuning with different hyper-parameters, and even the random seed that initiates the computation. Each layout provides a different lens on the interpoint distances in the data. The key structures in the high-dimensional data that we might hypothesize from these 2- D layouts are: (1) highly *separated clusters* (a, b, e, g, h) with the number of clusters ranging from 3-6; (2) *stringy branches* (f), and (3) *barely separated clusters* (c, d). They are at odds with each other, and it begs us to decide which are actually present in the p - D data.

The alternative approach to visualizing the high-dimensional data is to use linear projections, such as PCA. Tours, defined by [Asimov \(1985\)](#), broaden the scope by providing movies of linear projections showing views of the data from all directions. [Lee et al. \(2021\)](#) has a review of tour methods. Many tour algorithms are available in the R package `tourr` ([Wickham et al. 2011](#)), and versions enabling better interactivity in `langevitour` ([Harrison 2023](#)) and `detourr` ([Hart & Wang 2022](#)). Linear projections are a safe way to view high-dimensional data because they do not warp the space, so they are more faithful representations of the structure. However, linear projections can be cluttered, and global patterns can obscure local structure. The simple activity of projecting data from p - D suffers from piling ([Laa et al. 2022](#)), where data concentrates in the center of projections. NLDR is designed to escape these issues, to exaggerate structure so that it can be observed. But as a result, NLDR can hallucinate wildly, to suggest patterns that are not actually present in the data. [Lee et al. \(2022\)](#) provided interactive graphics to link between the NLDR view and tours, to help understand how the NLDR transformed the interpoint distances into a p - D layout.

Here, we further develop this approach to directly use the tour to examine how the NLDR is warping the space. It follows what [Wickham et al. \(2015\)](#) describes as **model-in-the-data-space**. The fitted model should be overlaid on the data to examine the fit relative to the spread of the observations. While this is straightforward, and commonly done when data is 2- D , it is also possible in p - D , for many models, when a tour is used.

[Wickham et al. \(2015\)](#) provides several examples of models overlaid on the data in p - D . In hierarchical clustering, a representation of the dendrogram using points and lines can be constructed by augmenting the data with points marking the merging of clusters. Showing the movie of linear projections shows how the algorithm sequentially fitted the cluster model to the data. For linear discriminant analysis or model-based clustering, the model can be indicated by $(p - 1)$ - D ellipses. It is possible to see whether the elliptical shapes appropriately match the variance of the relevant clusters, and to compare and contrast different fits. For PCA, one can display the model (a k - D plane of the reduced dimension) using wireframes of transformed cubes. Using a wireframe is the approach we take here to represent the NLDR model in p - D .

3 Considering the NLDR layout as a model

At first glance, thinking of NLDR as a modeling technique might seem strange. It is a simplified representation or abstraction of a system, process, or phenomenon in the real world. The p - D observations are the realization of the phenomenon, and the k - D NLDR layout is the simplified representation. Typically, $k = 2$ is used. From a statistical perspective, we can consider the distances between points in the 2- D layout to be the

variance that the model explains, and the (relative) difference with their distances in p - D as the error, or unexplained variance. We can also imagine that the positioning of points in 2- D represents the fitted values, which will have some prescribed position in p - D that can be compared with their observed values. This is the conceptual framework underlying the more formal versions of factor analysis (Jöreskog 1969) and MDS. (Note that, for this thinking, the full p - D data needs to be available, not just the interpoint distances.)

We define an NLDR method as taking as input an $n \times p$ matrix $\mathbf{X}$, as well as user-specified hyper-parameters $\boldsymbol{\theta}$, and returning an $n \times 2$ matrix $\mathbf{Y}$. We will refer to rows i of $\mathbf{X}$ as $\mathbf{x}_{i*}$ and columns j as $\mathbf{x}_{*j}$, and similarly for other matrices. Conceptually, we will regard an NLDR method as first “fitting” a function $g: \mathbb{R}^p \rightarrow \mathbb{R}^2$, and then calculating the rows of the output matrix using $\mathbf{y}_{i*} = g(\mathbf{x}_{i*})$. The implementation of an NLDR method may not actually divide its working into these two steps. Some but not all NLDR method implementations provide access to an estimated g , allowing supplementary points in p - D to be mapped to 2- D .

With our goal being to make a representation of this 2- D layout that can be lifted into high-dimensional space, the layout needs to be augmented to include neighbor information. A simple approach would be to triangulate the points and add edges. A more stable approach is to first bin the data, reducing it from n to $m \leq n$ observations, and connect the bin centroids. We recommend using a hexagon grid because it better reflects the data distribution and has fewer artifacts than a rectangular grid (Carr et al. 1987). This process serves to reduce some noisiness in the resulting surface shown in p - D . The steps in this process are shown in Figure 2, and documented below.

To illustrate the method and how to use it to choose from a selection of layouts, we simulated

7- D data, which we call the “2NC7” data, that has two separated nonlinear clusters, one forming a 2- D curved shape (cluster 1), and the other a 3- D curved shape (cluster 2), each consisting of 1000 observations. The first four variables, $\mathbf{x}_{*1}, \mathbf{x}_{*2}, \mathbf{x}_{*3}, \mathbf{x}_{*4}$ hold this cluster structure, and the remaining three $\mathbf{x}_{*5}, \mathbf{x}_{*6}, \mathbf{x}_{*7}$ are low-variance noise. Because no error was simulated in the first four variables, we would consider $[\mathbf{x}_{*1}, \mathbf{x}_{*2}, \mathbf{x}_{*3}, \mathbf{x}_{*4}]$ to be **the geometric structure (true model)** that we hope to capture. In practice, all variables might have some small error, and a structure to detect might be in any or all variables. The 2NC7 data is simple but complex enough to illustrate the new methodology. We would consider the two separated clusters as *global structure*, and the nonlinear surfaces to be relatively *local structure*. Figure 2 uses a **reasonable 2- D NLDR layout** of 2NC7, produced by tSNE, which reveals the two clusters with moderate separation (global), and flattens the curvilinear forms while (hopefully) preserving the proximity of points (local). Note that, because cluster 2 is 3- D , it cannot be fully captured in a 2- D layout, and the flat rectangle is as good as can be expected.

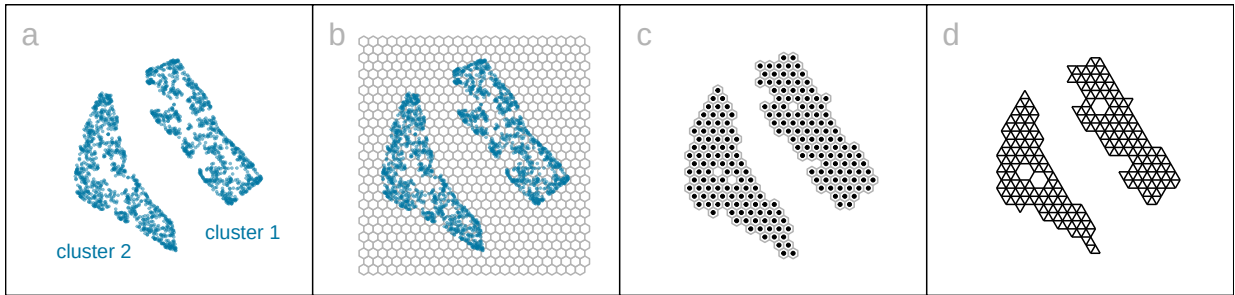


Figure 2: Key steps for constructing the model on the tSNE layout ($k = 2$): (a) data, (b) hexagon bins, (c) bin centroids, and (d) triangulated centroids. The layout used is a reasonable representation of the 2NC7 data, which contains two curvilinear clusters that have been flattened into a triangular and rectangular shape.

4 Algorithm to represent the model in 2- D

4.1 Scale the layout data

Because the algorithm is based on distances between points, scaling the layout data to range between $[0, 1]$ in the first coordinate is recommended. By default, the aspect ratio produced by the NLDR $(r_1, r_2, \dots, r_k)$ is preserved in the other coordinates. When $k = 2$, as in hexagon binning, the default range for the second coordinate is $[0, r_2/r_1]$.

4.2 Hexagon grid configuration

Although there are several implementations available (e.g., Carr et al. (2023)), the full set of data objects is needed for this project. So we have re-implemented the method so that all items needed for creating the p - D representation are available (illustrated in Figure 3).

The 2- D hexagon grid is uniquely specified by the hexagon ids, $h(= 1, \dots, b)$, and their 2- D centroids, $\mathbf{c}_{h*}^{(2)} = (c_{h1}, c_{h2})$. The algorithm inputs b_1 , which determines b_2 using the NLDR ratio, producing $b = b_1 \times b_2$, the total number of bins in the grid.

Buffer is used to ensure that the grid covers the range of data values, and provides flexibility for where the grid positining. It is well-known that where a grid starts can change binning results, even though most documentation focuses on binwidth. The buffer (q) is set as a proportion of the range. By default, $q = 0.1$, which is 10% of the range. The lower left position where the grid starts is set to be $(s_1 = -q, s_2 = -qr_2)$, specifying the centroid of the lower left hexagon, $\mathbf{c}_{1*}^{(2)} = (c_{11}, c_{12})$.

Considering the upper bound of the first NLDR component, the horizontal binwidth must

satisfy $a_1 > (1 + 2q)/(b_1 - 1)$. Similarly, the vertical binwidth must satisfy

$$a_2 \geq \frac{r_2 + q(1 + r_2)}{(b_2 - 1)}. \quad (1)$$

Because we are using regular hexagons $a_2 = \sqrt{3}a_1/2$, and Equation 1 can be re-written as

$$a_1 \geq \frac{2(r_2 + q(1 + r_2))}{\sqrt{3}(b_2 - 1)}.$$

This is a linear optimization problem, resulting in calculating b_2 as

$$b_2 = \left\lceil 1 + \frac{2(r_2 + q(1 + r_2))(b_1 - 1)}{\sqrt{3}(1 + 2q)} \right\rceil.$$

where $\lceil \cdot \rceil$ indicates “smallest integer greater than or equal to”.

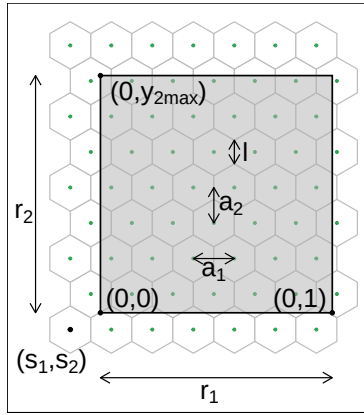


Figure 3: The components of the hexagon grid illustrating notation. The grey region indicates the extent of the data, which the grid must cover.

4.3 Binning the data

Observations are grouped into bins based on their nearest centroid. This produces a reduction in size of the data from n to m , where m (number of non-empty bins) $\leq b$ (total number of bins). Each point i is assigned to a bin using

$$u_i = \arg \min_{h=1, \dots, b} \sqrt{(y_{i1} - c_{h1}^{(2)})^2 + (y_{i2} - c_{h2}^{(2)})^2}$$

and then collected into sets $H_h = \{i | u_i = h\}$ ($h = 1, \dots, b$). Then $\hat{\mathbf{y}}_{i*} = \mathbf{c}_{u_i*}^{(2)}$ ($i = 1, \dots, n$) are considered the fitted values, from which we can also calculate 2- D residuals. The bin ID, u_i , for each observation is also needed to construct the model representation in high dimensions.

4.4 Indicating neighborhood

Delaunay triangulation (Lee & Schachter 1980, Gebhardt et al. 2024) is used to connect centroids to indicate neighboring observations in the NLDR layout (Figure 2 (d)). These are retained to preserve the neighborhood information when the model is lifted into p - D .

5 The model in p - D

5.1 Lifting from 2- D to p - D

The next step is to lift the 2- D model into p - D by computing p - D vectors that represent bin centroids. We use the p - D mean of the points in a given hexagon, H_h , denoted $\mathbf{c}_{h*}^{(p)}$,

to map the centroid $\mathbf{c}_{h*}^{(2)} = (c_{h1}, c_{h2})$ to a point in p - D . Let the j^{th} component of the p - D mean be

$$c_{hj}^{(p)} = \frac{1}{n_h} \sum_{i \in H_h} x_{ij}, \quad h = 1, \dots, b; \quad j = 1, \dots, p; \quad n_h > 0.$$

Then $\hat{\mathbf{x}}_{i*} = \mathbf{c}_{u_i}^{(p)}$ ($i = 1, \dots, n$) are considered the fitted values, from which we can also calculate p - D residuals. We could conceptualize this as a pseudo-inverse g^- , because for any 2- D observation, $\mathbf{y}_{i*}$ in a non-empty bin, we can provide coordinates for it in p - D , which would correspond to $\hat{\mathbf{x}}_{i*}$.

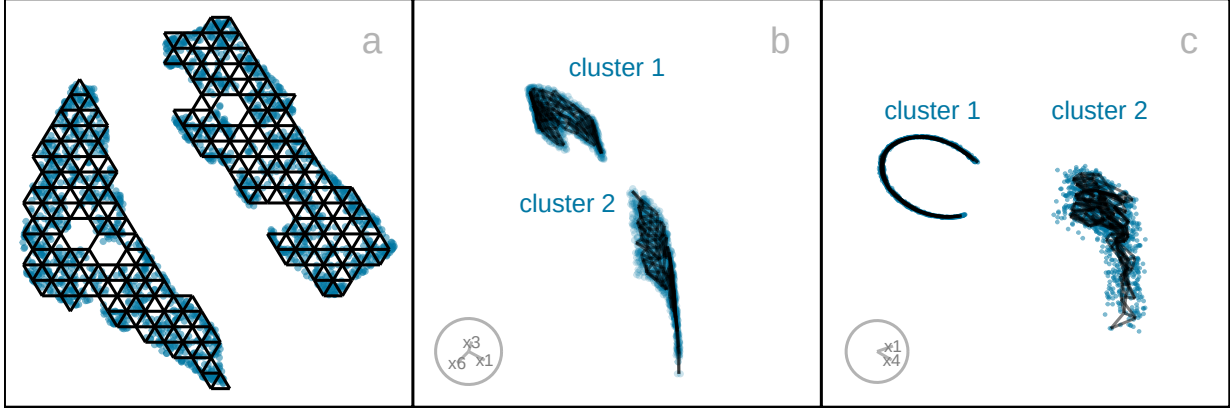


Figure 4: Lifting the 2- D fitted model (a) into p - D (b, c). Two projections of the p - D fitted model overlaying the data are shown in b, c. The fit is reasonably tight with the data in cluster 1, but slightly less so in cluster 2, probably because it is 3- D . Notice also that, in the 2- D layout, the two clusters have internal gaps which create a model with some holes. This lacy pattern happens regardless of the hyper-parameter choice for tSNE, but this doesn't severely impact the p - D model representation.

5.2 Measuring the fit

All NLDR methods internally optimize an objective function to produce a layout for a given set of hyper-parameters. These objective functions are not always made available to the user in the model output and or comparable across methods.

A range of established metrics is commonly used to assess the quality of NLDR embeddings, each emphasizing different aspects of structure preservation. For example, the *RNX* curve summarizes neighborhood agreement between p - D and k - D spaces across a range of scales, with its area under the curve (ARNX) reflecting a balance between local and global structure preservation (Lee et al. 2015). Random Triplet Accuracy (RTA) evaluates the consistency of distance orderings among triplets of points in 2 - D and p - D spaces, providing a measure of geometric preservation (Wang et al. 2021). The Shepard diagram (Shepard 1962), together with its associated Spearman correlation (SC) (Spearman 1961), assesses the monotonic relationship between pairwise distances in the two spaces and is often interpreted as a measure of global structure preservation. The Global Score (GS) compares the embedding to a PCA baseline and evaluates how well the overall geometry is retained (Amid & Warmuth 2022).

These metrics provide useful but complementary perspectives, as they capture different and often competing objectives, such as preserving local neighborhoods versus global distances. As a result, it is common for these measures to rank embeddings differently, reflecting the multi-objective nature of NLDR rather than a deficiency of any single metric.

To facilitate comparison with our proposed measure, we use reversed forms of these metrics (rRTA, rSC, rGS, and rARNX), so that lower values consistently indicate better performance.

While the above measures evaluate properties of the embedding itself, our goal is different: to assess how well the embedding supports a representation of the data in the original p - D space. This motivates a complementary approach based on model residuals.

We treat the embedding as defining a fitted representation in p - D . Each observation $\mathbf{x}_{i*}$ has an associated bin u_i with a centroid that serves as its fitted value, $\widehat{\mathbf{x}}_{i*} = \mathbf{c}_{u_i*}^{(p)}$. The

discrepancy between observed and fitted values can then be interpreted as a residual.

We define the hexbin error (HBE) as the root mean squared Euclidean distance between observations and their associated bin centroids:

$$\text{HBE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p (x_{ij} - \hat{x}_{ij})^2}. \quad (2)$$

We can consider $e_i = \sqrt{\sum_{j=1}^p (x_{ij} - \hat{x}_{ij})^2}$ to be the residual for each observation. Figure 5 visualizes these residuals as a density (a), as colors in the NLDR layout (b), and in a tour (c). Larger residuals are concentrated in one cluster, reflecting the intentional design that this cluster has an intrinsic higher-dimensional structure (approximately 3- D) and therefore cannot be fully represented in a 2- D embedding. Importantly, HBE is not intended to replace existing embedding-quality metrics. Instead, it provides a complementary diagnostic perspective, highlighting where the embedding fails to represent the original data structure in p - D . Differences between HBE and other metrics can therefore be interpreted as informative signals about trade-offs between local and global structure preservation.

5.3 Diagnosing the fit with interactive graphics

Matching points in the 2- D layout with their positions in p - D is useful when assessing the fit, deciding on a layout, and basically understanding what a layout communicates about the data. The interactive 2- D layout (Sievert 2020) and the langevitour (Harrison 2023) view with the fitted model overlaid can be linked using a browsable HTML widget (Cheng & Sievert (2025), Cheng et al. (2024)). A rectangular “brush” is used to select points in one plot, which will highlight the corresponding points in the other plot(s). Because the

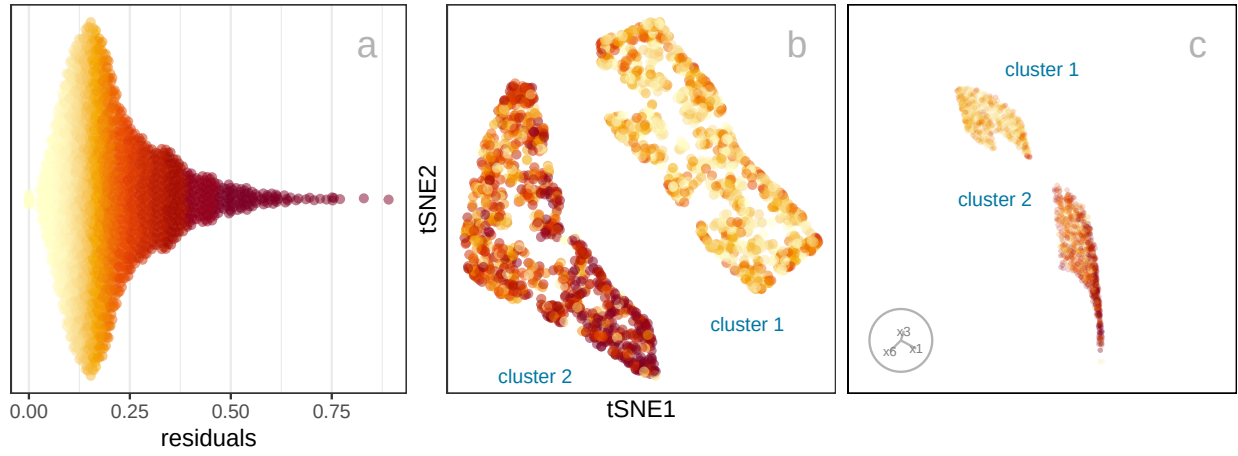


Figure 5: Examining the distribution of residuals in a jittered dotplot (a), 2- D NLDR layout (b), and a tour of 4- D data space (c). Color indicates residual value (e_i), dark indicating high value. Most large residuals are distributed in cluster 2, which is due to the 2- D 's inability to capture the 3- D structure.

langevitour is dynamic, brush events that become active will pause the animation, so that a user can interrogate the current view. This approach is illustrated in the examples in the Appendix and video supplements, to show how it can help to understand how an NLDR has organized the observations.

6 Choosing from a selection of layouts

Figure 6 shows the diagnostic plot of HBE vs binwidth (i) for a selection of plots from which to use. Color in all plots indicates the layout (a-f). The details of how they were generated are in the Appendix. There is nothing special about this selection of layouts other than that all five methods were used: (a, f) tSNE, (b) UMAP, (c) PHATE, (d) TriMAP, (e) PaCMAP. They represent different patterns, but all agree on there being two clusters, with varying distance and shape between them. The process to make a decision is:

1. Fit the model to a range of binwidths for each layout, calculating HBE.

2. Plot each HBE against binwidth, for all layouts.
3. The better representation(s) will have universally lower HBE. When the lines cross, one needs to decide whether the local structure (smaller binwidth) is more important than the larger structure (large binwidth).
4. Examine the model for the chosen layout in p - D . The exact choice of binwidth isn't important, but we recommend a reasonably small one, so that the wireframe is smoother, and there is a chance to see small details of departures between the fitted model and the data. (See Figure 4 b, c for this data.)

For this set of layouts, either a or f would be preferable to the other layouts, because the HBE values are lower at every binwidth. (Note that a was used to illustrate the approach in the previous sections.)

Using binwidth as the horizontal axis is a choice. An alternative is to use the average bin count. In theory, this is the more sensible choice. Because it summarizes the grid occupancy of the data. Using binwidth has the problem that a smaller binwidth for one layout might be equivalent to a larger binwidth in a different layout, as a fraction of the observations. Using binwidth sets up a possible mismatching HBE value. However, using average bin count tends to lose the ability to penalize layouts that exaggerate inter-cluster distance (see plots in the Appendix). An argument for why binwidth is the better choice is that it treats the 2 - D space as a uniform canvas on which to compare the representations.

It can also be useful to calculate the suite of other metrics to measure the fit, for each layout, which is done in plot ii. The comparison with HBE requires that a single value be chosen. Here, $a_1 = 0.05$ is used, but the actual choice wouldn't substantially matter

because the ordering of the curves is quite consistent across binwidths. Note that in this parallel coordinate plot, the individual metrics are individually scaled to range between 0 and 1. HBE shows most agreement with rARNX, both rank a and f the best. They differ in how they rate (1) layout d - HBE would consider it the worst, and rARNX to be not quite as good as a and f - and (2) layout c - rARNX considers it to be the worst, and HBE in the middle. The other three metrics roughly agree with each other and are almost the polar opposite of HBE. This is most likely explained by the tension between local and global structure. Most will prefer a bigger distance between clusters in the layout, or at least not penalize this. The problem is that this leads to incorrect perception when the clusters are not widely separated, as is the case for this data (Figure 4).

The final check is to examine the chosen model fit (or several top-rated fits) in the p - D data space Figure 4 (b, c). As with all model fitting, if the data closely hugs the model representation, in all projections shown by the tour, then the choice is further supported. This is a suitable approach to better understand why metrics disagree on layouts. The Appendix has a comparison of the model fits for layouts a and c. Layout c twists extremely along the 2- and 3- D manifolds where the data lies. Twisting in the fitted model appears to be very common for all methods for many data sets, which one could imagine as a solution for squashing higher dimensions into lower ones while preserving the neighborhood. In this case, it is quite extreme.

You might find that some elements of the p - D structure are captured, but others are not. If this happens, a strategy is to break the data into subsets and repeat the process of deciding on a layout. Showing multiple layouts for a data set may better illustrate its nature. This might be advisable with 2NC7 also. At first glance, layout b looks preferable, because the

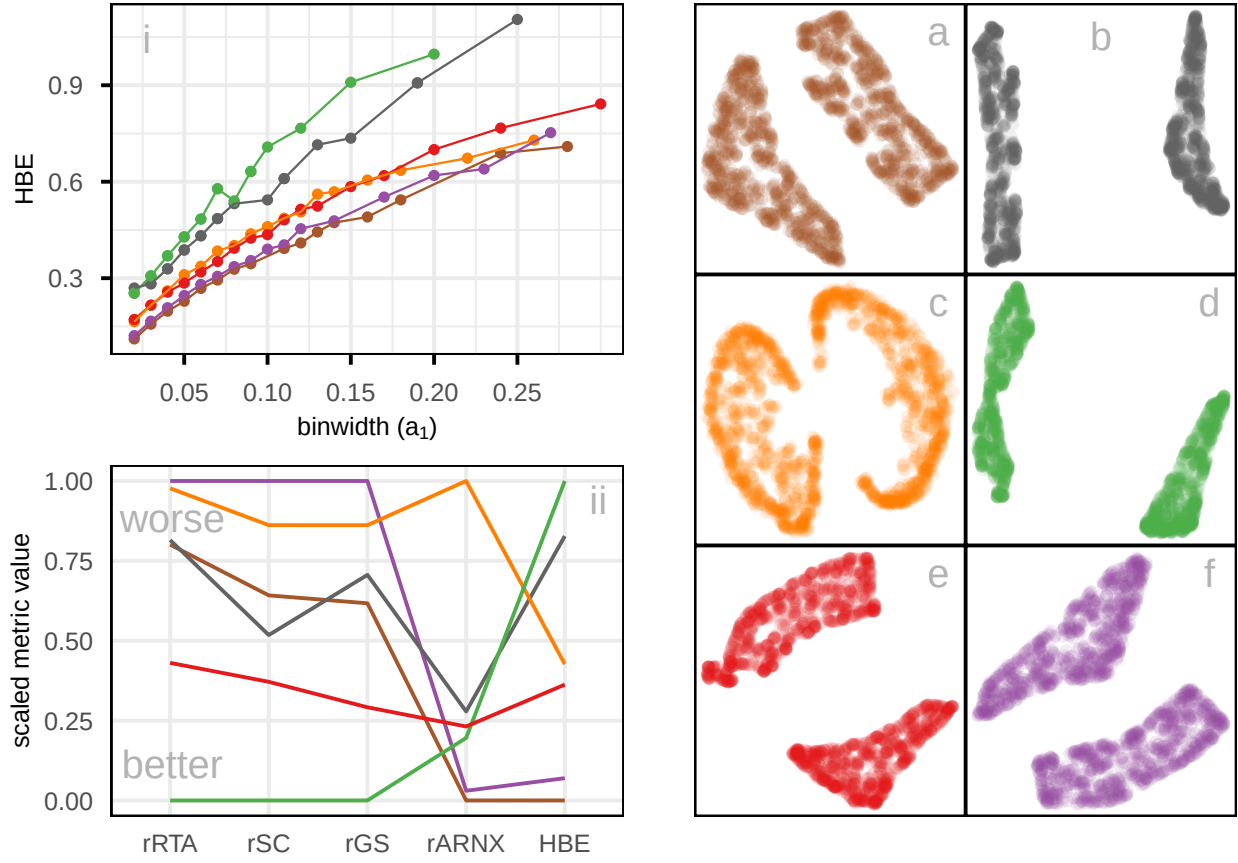


Figure 6: Assessing which of the 6 NLDR layouts (a-f) on the 2NC7 data is the better representation using (i) HBE for varying binwidth (a_1), and (ii) scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.05$). Color represents the NLDR layout. Using HBE (i), layouts a and f are nearly indistinguishable from each other but have universally lower values than the others: either layout could adequately represent the data. HBE mostly agrees with rARNX, except for layout d. The other metrics rank the layouts in reverse order! The problem, though, is that they favor more separation, which is misleading for this data. This can be validated by viewing the data in a tour.

distribution of observations within each cluster is smoother than in either (a) or (f). This likely reflects that the layout better captures the local structure.

7 Prediction into 2-D

Only UMAP provides a way to predict the layout positions of new observations ([Konopka 2023](#)) when the full data is available, and this is made available in the `umap` ([McInnes](#)

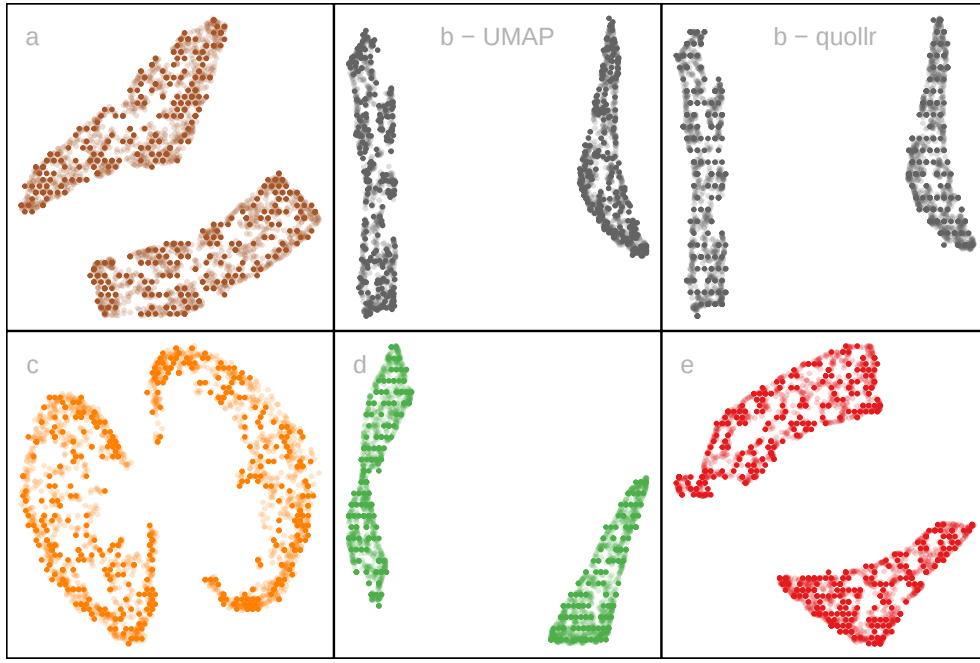


Figure 7: Predicted test data (dark) overlaid on training sets (light), for (most) layouts shown in Figure 6. UMAP is the only method to provide predicted test set locations, and these are compared with our approach in panels (b - UMAP) and (b - quollr). All other panels show predictions obtained using our method applied. The predicted test set data match the training set layouts very neatly.

et al. 2018) software. Other approaches for embedding new data include neural network autoencoders (Hinton & Salakhutdinov 2006) and parametric tSNE (van der Maaten 2009) but they do not directly relate to the common methods presented here.

A benefit of our approach is that with the model representation, we can predict layout positions for any method. To illustrate this, we will first compare the approach with the predictions provided by UMAP, by generating a test set of 2NC7. We will also show predictions of this test set for the layouts used in Figure 6.

Our procedure for the predicting the layout position of a new observation is: (i) identify the closest bin centroid in p -D, $\mathbf{c}_{h*}^{(p)}$, and (ii) assign the corresponding centroid in 2-D, $\mathbf{c}_{h*}^{(2)}$, as the predicted embedding. This is not entirely different from how UMAP provides predictions, which is to find the nearest neighbors in p -D, and interpolate their positions in

2- D to provide the predicted embedding. This can be considered to be an estimate of the mapping g that maps the new observation $(\mathbf{x}^*)$ into the layout $(\mathbf{y}^*)$, $\mathbf{y}^* = \hat{g}(\mathbf{x}^*)$, $\hat{g}: \mathbb{R}^p \rightarrow \mathbb{R}^2$.

Figure 7 shows the results for UMAP (a) and our predictions (b). The 2NC7 data is divided into training and test sets, using a 70% split. The training data is used to make the 2- D layout. (This is similar but not quite the same as the layout on the full data seen in Figure 6 plot b.) The test set is held out for prediction. In both cases, it can be seen that the test set matches the training set in the layout quite well. The main differences are that our predictions are to the bin centroid, so it looks gridded. Adding some jitter to the predicted values would make them look more similar to the UMAP predictions. Just because we can, doesn't mean we should - for this data, the method provides demonstrably reasonable results, and better than having no method at all.

8 Applications

To illustrate the approach, we use two examples: PBMC3k data (single cell gene expression), where an NLDR layout is used to represent cluster structure present in the p - D data, and MNIST hand-written digits, where NLDR is used to represent a low-dimensional nonlinear manifold in p - D .

8.1 PBMC3k

This is a benchmark single-cell RNA-Seq data set collected on Human Peripheral Blood Mononuclear Cells (PBMC3k) as used in [10x Genomics \(2016\)](#). Single-cell data measures the gene expression of individual cells in a sample of tissue (see, for example, [Haque et al.](#)

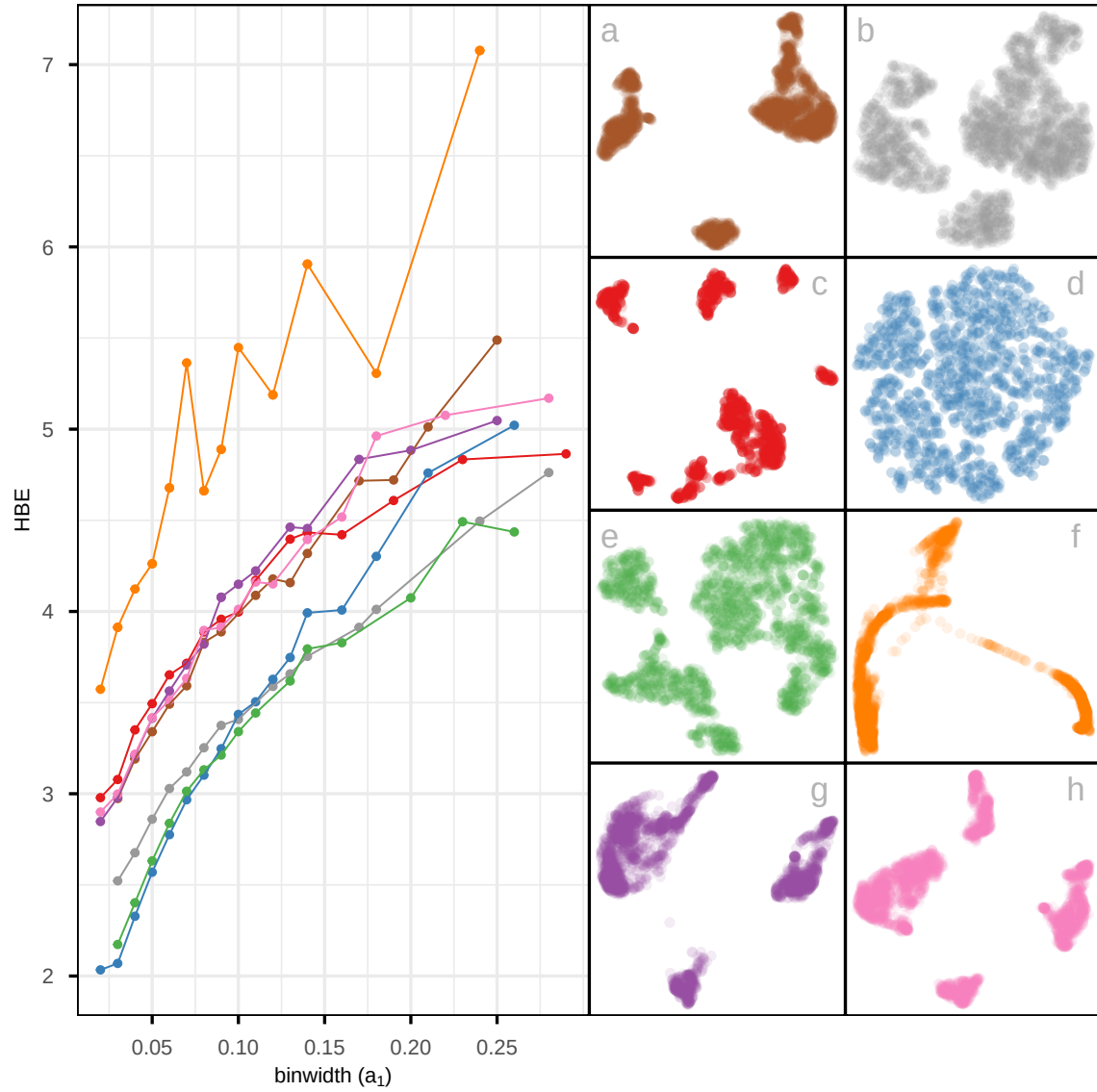


Figure 8: Assessing which of the 8 NLDR layouts on the PBMC3k data (shown in Figure 1) is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color used for the lines and points in the left plot and in the scatterplots represents the NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, and h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of the best is between layouts b and e, which have small separations between oddly shaped clusters. Layout e is chosen as the best. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

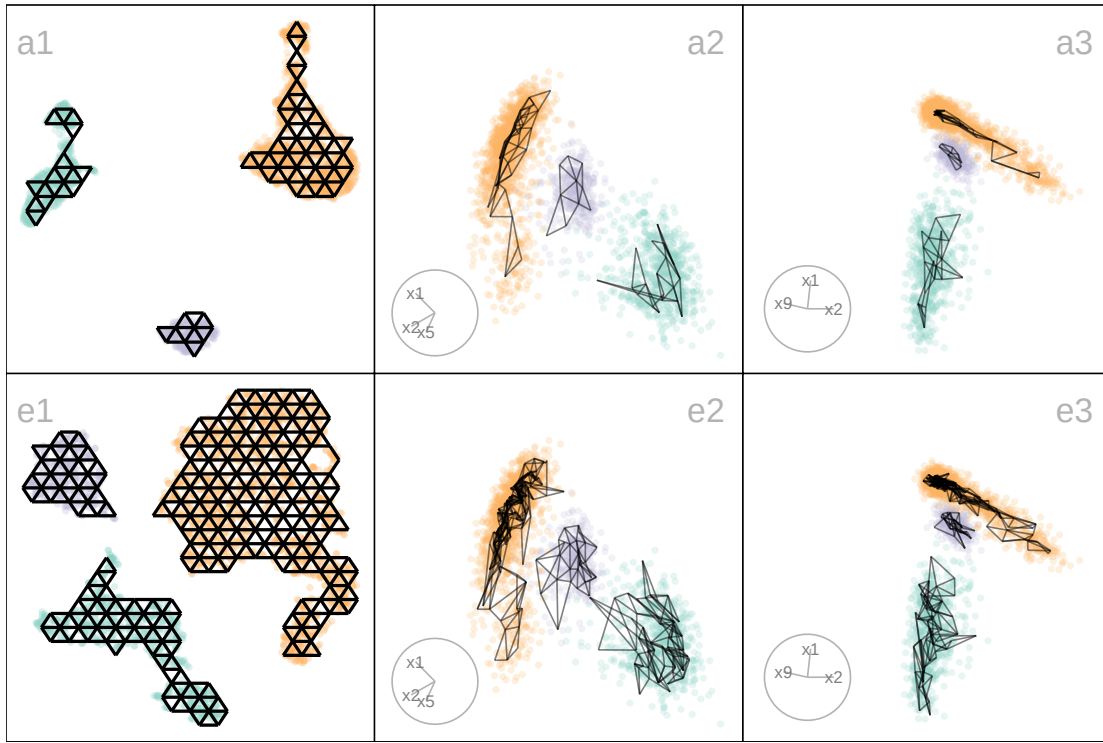


Figure 9: Compare the published 2- D layout (a) made with UMAP and the 2- D layout selected by HBE plot (e) made by tSNE. The two plots on the right show projections from a tour, with the models overlaid. The published layout suggests three very separated clusters, but this is not present in the data. While there may be three clusters, they are not well-separated. The difference in model fit also indicates this: the published layout a does not spread out fully into the point cloud like the model generated from layout e. This supports the choice that layout e is the better representation of the data, because it does not exaggerate separation between clusters.

(2017)). This type of data is used to obtain an understanding of cellular-level behavior and heterogeneity in their activity. Clustering of single-cell data is used to identify groups of cells with similar expression profiles. NLDR is often used to summarize the cluster structure. Usually, NLDR does not use the cluster labels to compute the layout, but uses color to represent the cluster labels when it is plotted.

In this data, there are 2622 single cells and 1000 gene expressions (variables). Following the same pre-processing as [Chen et al. \(2024\)](#), different NLDR techniques were performed on the first nine principal components. Figure 1 shows this data using a variety of methods

and different hyper-parameters. You can see that the result is wildly different depending on the choices. Layout a is a reproduction of the layout that was published in [Chen et al. \(2024\)](#). This layout suggests that the data has three very well-separated clusters, each with an odd shape. The question is whether this accurately represents the cluster structure in the data, or whether they should have chosen b, c, d, e, f, g, or h. This is what our new method can help with – to decide which is the more accurate 2- D representation of the cluster structure in the p - D data.

Figure 8 shows HBE across a range of binwidths (a_1) for each of the layouts in Figure 1. The layouts were generated using tSNE and UMAP with various hyper-parameter settings, while PHATE, PaCMAP, and TriMAP were applied using their default settings. Lines are color-coded to match the color of the layouts shown on the right. Lower HBE indicates a better fit. Using a range of binwidths shows how the model changes, with possibly the best model being one that is universally low HBE across all binwidths. It can be seen that layout f is sub-optimal with universally higher HBE. Layout a, the published one, is better, but it is not as good as layouts b, d, or e. With some imagination, layout d perhaps shows three barely distinguishable clusters. Layout e shows three, possibly four, clusters that are more separated. The choice reduces from eight to these two. Layout d has slightly better HBE when the a_1 is small, but layout e beats it at larger values. Thus, we could argue that layout e is the most accurate representation of the cluster structure of these eight.

To further assess the choices, we need to look at the model in the data space, by using a tour to show the wireframe model overlaid on the data in the 9- D space (Figure 9). Here we compare the published layout (a) versus what we argue is the best layout (e). The top row (a1, a2, a3) corresponds to the published layout, and the bottom row (e1, e2, e3)

corresponds to the optimal choice according to our procedure. The middle and right plots show two projections. The primary difference between the two models is that the model of layout e does not fill out to the extent of the data but concentrates in the center of each point cloud. Both suggest that three clusters are a reasonable interpretation of the structure, but the layout e more accurately reflects the separation between them, which is small.

8.2 MNIST hand-written digits

The digit “1” of the MNIST dataset (LeCun et al. 1998) consists of 7877 grayscale images of handwritten “1”s. Each image is 28×28 pixels, which corresponds to 784 variables. The first 10 principal components, explaining 83% of the total variation, are used. This data essentially lies on a nonlinear manifold in high dimensions, defined by the shapes that “1”s make when sketched. We expect that the best layout captures this type of structure and does not exhibit distinct clusters.

Figure 10 compares the fit of six layouts computed using UMAP (b), PHATE (c), TriMAP (d), PaCMAP (e), with default hyper-parameter setting and two tSNE runs, one with default hyper-parameter setting (a) and the other changing perplexity to 89 (f). The layouts are reasonably similar in that they all have the observations in a single blob. Some (b, c) have a more curved shape than others. Layout e is the most different, having a linear shape and a single very large outlier. Both a and f have a small clump of points, perhaps slightly disconnected from the other points, in the lower to middle right.

The layout plots are colored to match the lines in the HBE vs binwidth (a_1) plot. Layouts

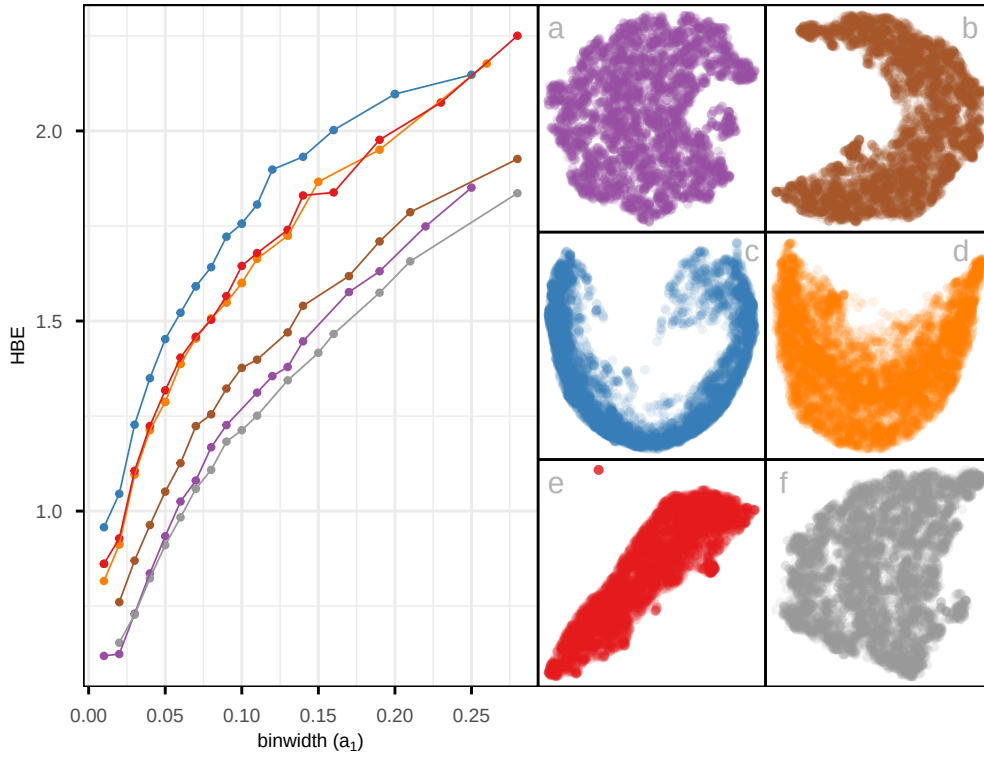


Figure 10: Assessing which of the 6 NLDR layouts of the MNIST digit 1 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color is used for the lines and points in the left plot to match the scatterplots of the NLDR layouts (a-f). Layout c is universally poor. Layouts a and f that show a big cluster and a small circular cluster are universally optimal. Layout a performs well at tiny binwidth (where most points are in their own bin) and not as well as f with larger binwidth, thus layout f is the best choice. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

a, b, and f fit the data better than c, d, e, and layout f appears to be the best fit. Figure 11 shows this model in the data space in two projections from a tour. The data is curved in the 10- D space, and the fitted model captures this curve. The small clump of points in the 2- D layout is highlighted in both displays. These are almost all inside the curve of the bulk of points and are sparsely located. The fact that they are packed together in the 2- D layout is likely due to the handling of density differences by the NLDR.

An interesting aside is that the rather strange layout e, which has what looks like a single point far from the remaining observations, is actually similar to this one. That point is

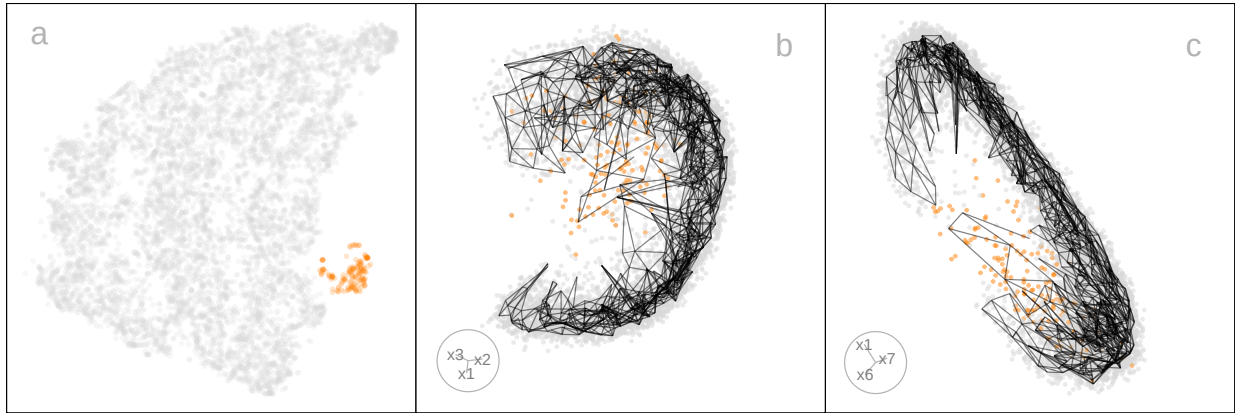


Figure 11: Summary from exploring tSNE layout of the MNIST digit 1 data (a in Figure 12) using linked brushing. There is a big nonlinear cluster (grey) and a small cluster (orange) located very close to one corner of the big cluster in 2- D (a). The MNIST digit 1 data has a nonlinear structure in 10- D . Two 2- D projections from a tour on 10- D reveal that the small orange cluster is actually a diffuse set of points wrapped within the grey cluster, which is C-shaped in the high dimensions.

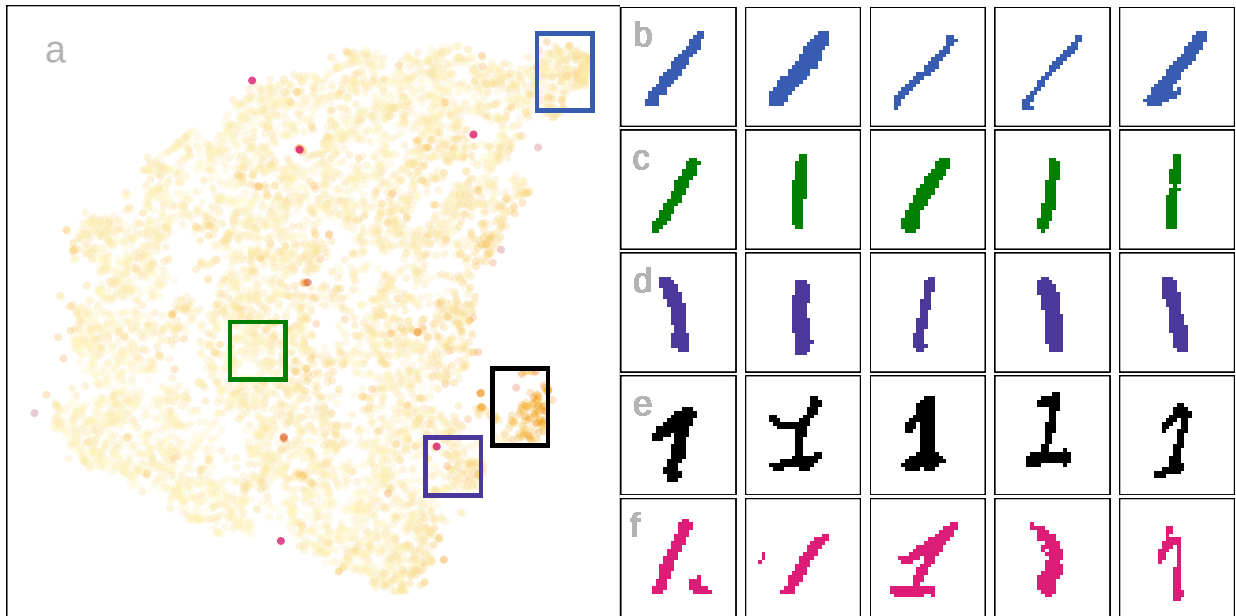


Figure 12: Summary of the layout structure, and large errors, relative to the MNIST digit 1 shape: (a) layout colored by residual value, and at right (b-e) are images of samples of observations taken at locations around the layout, showing similarity in how the 1's were drawn. Set (f) are images corresponding to large residuals in the big cluster (darker pink in plot a). Along the big cluster, the angle of digit 1 changes (b-d). The small cluster has larger residuals, and the images show that these tend to be European style with a flag at the top and a base at the bottom. The set in (f) shows various poorly written digits.

actually a clump of points corresponding to some of the diffuse points interior to the curve of the bulk of points. This is easy to see using the linked brushing tool.

The next step is to investigate the $2-D$ layout to understand what information is learned from this representation. Figure 12 summarizes this investigation. Plot a shows the layout with points colored by their residual value - darker color indicates larger residual and poor fit. The plots b, c, d, e show samples of hand-written digits taken from inside the colored boxes. Going from top to bottom around the curve shape, we can see that the “1”s are drawn from right slant to a left slant. The “1”s in d (black box) tend to have the extra up stroke, but are quite varied in appearance. The “1”s shown in the plots labelled e correspond to points with big residuals. They can be seen to be more strangely drawn than the others. Overall, this $2-D$ layout shows a useful way to summarize the variation in ways “1”s are drawn.

9 Discussion

We have developed an approach to help assess and compare NLDR layouts. It depends on conceptualizing the $2-D$ layout as a model, allowing for the creation of a wireframe representation of the model that can be lifted into $p-D$. The goodness of fit of the layout is measured by a new statistic, HBE, which can be used to compare multiple layouts. Global and local preservation of structure is assessed by examining the HBE across a range of binwidths. Viewing the fitted model in the high-dimensional data space can help to determine how well it captures structures in the data. This approach also provides a simple way to predict the values of new $p-D$ observations in the $2-D$, which could be useful for implementing uncertainty checks, such as using training and testing samples.

This methodology is made available in the R package `quollr`. The package has methods to

fit the model, compute diagnostics, and also visualize the results, with interactivity. We have primarily used the `langevitour` software (Harrison 2023) to view the model in the data space, but other tour software such as `tourr` (Wickham et al. 2011) and `detourr` (Hart & Wang 2022) could also be used.

Two examples illustrating usage are provided: the PBMC3k data, where the NLDR is summarizing clustering in p - D , and MNIST data illustrating how NLDR represents an intrinsically low-dimensional nonlinear manifold. We illustrated that a published UMAP layout of the PBMC3k dataset (Chen et al. 2024), is sub-optimal. The chosen layout inaccurately exaggerated separation between clusters, even suggesting separation where there was none. In the hand-written digits (LeCun et al. 1998), illustrating how an NLDR can condense some sparse structure, making it appear as a cluster. This also illustrated the use of model residuals to identify oddly written digits.

The supplementary material contains examples of strange behavior of NLDR methods, including twisting, flattened “pancake” clusters in p - D , and severe effects of density differences.

The HBE metric complements existing metrics for evaluating NLDR layout, ARNX, RTA, SC, and RGS. It suggests that more work on distance measures to summarize the fit could be valuable. HBE uses Euclidean distance, but other measures, such as geodesic distances (Tenenbaum et al. 2000), may help capture departures from curved or nonlinear surfaces in the data.

The approach presented here could also be extended for k - D ($k > 2$) layouts. Using 3- D binning, k -means clustering with nearest neighbors edge connection, or even special implementations of convex hulls, could provide model representations.

10 Supplementary Materials

All the materials to reproduce the paper can be found at <https://github.com/JayaniLakshika/paper-nldr-vis-algorithm>.

The supplementary materials provide additional details on the methods and hyperparameters used to generate layouts, video links of animated p - D tours, notation summaries, and the R and Python scripts used in the study. They also describe the generation of the 2NC7 data, computation of hexagon grid configurations, and data binning procedures. Further sections highlight interesting NLDR behaviors observed in the data space and compare HBE with existing evaluation metrics for the PBMC3k and MNIST datasets.

The R package `quollr`, available on CRAN and at <https://jayanilakshika.github.io/quollr/>, provides software accompanying this paper to fit the wireframe model representation, compute diagnostics, visualize the model in the data with `langevitour`, and link multiple plots interactively.

11 Acknowledgments

These R packages were used for the work: `tidyverse` (Wickham et al. 2019), `Rtsne` (Krijthe 2015), `umap` (Konopka 2023), `patchwork` (Pedersen 2024), `colorspace` (Zeileis et al. 2020), `langevitour` (Harrison 2023), `conflicted` (Wickham 2023), `reticulate` (Ushey et al. 2024), `kableExtra` (Zhu 2024). These python packages were used for the work: `trimap` (Amid & Warmuth 2022) and `pacmap` (Wang et al. 2021). The article was created with R packages `quarto` (Allaire & Dervieux 2024).

References

- 10x Genomics (2016), ‘3k PBMCs from a Healthy Donor, Universal 3’ Gene Expression dataset analysed using Cell Ranger 1.1.0’. Accessed: 2025-06-09. <https://www.10xgenomics.com/datasets/3-k-pbm-cs-from-a-healthy-donor-1-standard-1-1-0>.
- Allaire, J. & Dervieux, C. (2024), *quarto: R Interface to Quarto Markdown Publishing System*. R package version 1.4.4. <https://CRAN.R-project.org/package=quarto>.
- Amid, E. & Warmuth, M. K. (2022), ‘TriMap: Large-scale Dimensionality Reduction Using Triplets’, *arXiv preprint arXiv:1910.00204* .
- Asimov, D. (1985), ‘The Grand Tour: A Tool for Viewing Multidimensional Data’, *SIAM Journal of Scientific and Statistical Computing* **6**(1), 128–143.
- Borg, I. & Groenen, P. J. F. (2005), *Modern Multidimensional Scaling*, 2nd edn, Springer, New York, NY. <https://doi.org/10.1007/0-387-28981-X>.
- Carr, D. B., Littlefield, R. J., Nicholson, W. L. & Littlefield, J. S. (1987), ‘Scatterplot Matrix Techniques for Large N’, *Journal of the American Statistical Association* **82**(398), 424–436. <http://www.jstor.org/stable/2289444>.
- Carr, D., Lewin-Koh, N., Maechler, M. & Sarkar, D. (2023), *hexbin: Hexagonal Binning Routines*. R package version 1.28.3. <https://CRAN.R-project.org/package=hexbin>.
- Chari, T. & Pachter, L. (2023), ‘The specious art of single-cell genomics’, *PLoS Computational Biology* **19**, e1011288. <https://doi.org/10.1371/journal.pcbi.1011288>
- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type

- Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* **4**(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.
- Cheng, J. & Sievert, C. (2025), *crosstalk: Inter-Widget Interactivity for HTML Widgets*. R package version 1.2.2. <https://CRAN.R-project.org/package=crosstalk>
- Cheng, J., Sievert, C., Schloerke, B., Chang, W., Xie, Y. & Allen, J. (2024), *htmltools: Tools for HTML*. R package version 0.5.8.1. <https://CRAN.R-project.org/package=htmltools>
- Coifman, R. R., Lafon, S., Lee, A. B., Maggioni, M., Nadler, B., Warner, F. & Zucker, S. W. (2005), ‘Geometric Diffusions as a Tool for Harmonic Analysis and Structure Definition of Data: Diffusion Maps’, *Proceedings of the National Academy of Sciences of the United States of America* **102**(21), 7426–7431. <https://www.pnas.org/doi/abs/10.1073/pnas.0500334102>.
- Gebhardt, A., Bivand, R. & Sinclair, D. (2024), *interp: Interpolation Methods*. R package version 1.1-6. <https://CRAN.R-project.org/package=interp>.
- Haque, A., Engel, J., Teichmann, S. A. & Lönnberg, T. (2017), ‘A Practical Guide to Single-cell RNA-sequencing for Biomedical Research and Clinical Applications’, *Genome Medicine* **9**(1), 75. <https://doi.org/10.1186/s13073-017-0467-4>.
- Harrison, P. (2023), ‘langevitour: Smooth Interactive Touring of High Dimensions, Demonstrated With scRNA-Seq Data’, *The R Journal* **15**, 206–219. <https://doi.org/10.32614/RJ-2023-046>.
- Hart, C. & Wang, E. (2022), *detourr: Portable and Performant Tour Animations*. R package version 0.1.0. <https://casperhart.github.io/detourr/>.

- Hinton, G. E. & Salakhutdinov, R. R. (2006), ‘Reducing the dimensionality of data with neural networks’, *Science* **313**(5786), 504–507.
- Irizarry, R. (2024), ‘Biologists, stop putting umap plots in your papers’, <https://simplystatistics.org/posts/2024-12-23-biologists-stop-including-umap-plots-in-your-papers/>. Blog post on Simply Statistics.
- Johnstone, I. M. & Titterton, D. M. (2009), ‘Statistical Challenges of High-Dimensional Data’, *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **367**(1906), 4237–4253. <https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2009.0159>.
- Jolliffe, I. (2011), *Principal Component Analysis*, Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 1094–1096. https://doi.org/10.1007/978-3-642-04898-2_455.
- Jöreskog, K. G. (1969), ‘A General Approach to Confirmatory Maximum Likelihood Factor Analysis’, *Psychometrika* **34**(2), 183–202. <https://doi.org/10.1007/BF02289343>.
- Koch, I. (2013), *Analysis of Multivariate and High-Dimensional Data*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press.
- Konopka, T. (2023), *umap: Uniform Manifold Approximation and Projection*. R package version 0.2.10.0. <https://CRAN.R-project.org/package=umap>.
- Krijthe, J. H. (2015), *Rtsne: T-Distributed Stochastic Neighbor Embedding Using Barnes-Hut Implementation*. R package version 0.16. <https://github.com/jkrijthe/Rtsne>.
- Kruskal, J. B. (1964), ‘Nonmetric Multidimensional Scaling: A Numerical Method’, *Psychometrika* **29**(2), 115–129. <https://doi.org/10.1007/BF02289694>.

- Laa, U., Cook, D. & Lee, S. (2022), ‘Burning Sage: Reversing the Curse of Dimensionality in the Visualization of High-Dimensional Data’, *Journal of Computational and Graphical Statistics* **31**(1), 40–49. <https://doi.org/10.1080/10618600.2021.1963264>.
- LeCun, Y., Cortes, C. & Burges, C. J. C. (1998), ‘The MNIST Database of Handwritten Digits’. Accessed: 2025-06-09. <http://yann.lecun.com/exdb/mnist/>.
- Lee, D. T. & Schachter, B. J. (1980), ‘Two Algorithms For Constructing a Delaunay Triangulation’, *International Journal of Computer & Information Sciences* **9**(3), 219–242. <https://doi.org/10.1007/BF00977785>.
- Lee, J. A., Peluffo-Ordóñez, D. H. & Verleysen, M. (2015), ‘Multi-scale similarities in stochastic neighbour embedding: Reducing dimensionality while preserving both local and global structure’, *Neurocomputing* **169**, 246–261. <https://www.sciencedirect.com/science/article/pii/S0925231215003641>
- Lee, S., Cook, D., da Silva, N., Laa, U., Wang, E., Spyrisson, N. & Zhang, H. S. (2021), ‘A Review of the State-of-the-Art on Tours for Dynamic Visualization of High-Dimensional Data’, *arXiv preprint arXiv:2104.08016* .
- Lee, S., Laa, U. & Cook, D. (2022), ‘Casting multiple shadows: interactive data visualisation with tours and embeddings’, *Journal of Data Science, Statistics, and Visualisation* **2**(3). <https://jdssv.org/index.php/jdssv/article/view/21>
- Maaten, L. V. D. & Hinton, G. E. (2008), ‘Visualizing Data Using t-SNE’, *Journal of Machine Learning Research* **9**(11), 2579–2605. <https://api.semanticscholar.org/CorpusID:5855042>.
- McInnes, L., Healy, J., Saul, N. & Großberger, L. (2018), ‘UMAP: Uniform Manifold

- Approximation and Projection’, *Journal of Open Source Software* **3**(29), 861. <https://doi.org/10.21105/joss.00861>.
- Moon, K. R., van Dijk, D., Wang, Z., Gigante, S. A., Burkhardt, D. B., Chen, W. S., Yim, K., van den Elzen, A., Hirn, M. J., Coifman, R. R., Ivanova, N. B., Wolf, G. & Krishnaswamy, S. (2019), ‘Visualizing Structure and Transitions in High-Dimensional Biological Data’, *Nature Biotechnology* **37**, 1482–1492. <https://doi.org/10.1038/s41587-019-0336-3>.
- Pedersen, T. L. (2024), *patchwork: The Composer of Plots*. R package version 1.2.0. <https://CRAN.R-project.org/package=patchwork>.
- Saeed, N., Nam, H., Haq, M. I. U. & Muhammad Saqib, D. B. (2018), ‘A Survey on Multidimensional Scaling’, *ACM Comput. Surv.* **51**(3). <https://doi.org/10.1145/3178155>.
- Shepard, R. N. (1962), ‘The analysis of proximities: Multidimensional scaling with an unknown distance function. i.’, *Psychometrika* **27**(2), 125–140.
- Sievert, C. (2020), *Interactive Web-Based Data Visualization with R, plotly, and shiny*, Chapman and Hall/CRC. <https://plotly-r.com>
- Silva, V. & Tenenbaum, J. (2002), ‘Global Versus Local Methods in Nonlinear Dimensionality Reduction’, *Advances in Neural Information Processing Systems* **15**, 721–728. https://proceedings.neurips.cc/paper_files/paper/2002/file/5d6646aad9bcc0be55b2c82f69750387-Paper.pdf.
- Spearman, C. (1961), ‘The proof and measurement of association between two things.’
- Tenenbaum, J. B., de Silva, V. & Langford, J. C. (2000), ‘A global geometric framework

- for nonlinear dimensionality reduction’, *Science* **290**(5500), 2319–2323. <https://www.science.org/doi/abs/10.1126/science.290.5500.2319>
- Ushey, K., Allaire, J. & Tang, Y. (2024), *reticulate: Interface to Python*. R package version 1.38.0. <https://CRAN.R-project.org/package=reticulate>.
- van der Maaten, L. (2009), Learning a parametric embedding by preserving local structure, in ‘Artificial Intelligence and Statistics’, PMLR, pp. 384–391.
- Wang, Y., Huang, H., Rudin, C. & Shaposhnik, Y. (2021), ‘Understanding How Dimension Reduction Tools Work: An Empirical Approach to Deciphering t-SNE, UMAP, TriMap, and PaCMAP for Data Visualization’, *Journal of Machine Learning Research* **22**(201), 1–73. <http://jmlr.org/papers/v22/20-1061.html>.
- Wickham, H. (2023), *conflicted: An Alternative Conflict Resolution Strategy*. R package version 1.2.0. <https://CRAN.R-project.org/package=conflicted>.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), ‘Welcome to the Tidyverse’, *Journal of Open Source Software* **4**(43), 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, H., Cook, D. & Hofmann, H. (2015), ‘Visualizing Statistical Models: Removing the Blindfold’, *Statistical Analysis and Data Mining: The ASA Data Science Journal* **8**(4), 203–225. <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11271>.
- Wickham, H., Cook, D., Hofmann, H. & Buja, A. (2011), ‘tourr: An R Package For

Exploring Multivariate Data With Projections’, *Journal of Statistical Software* **40**(2), 1–18.
<http://www.jstatsoft.org/v40/i02/>.

Zeileis, A., Fisher, J. C., Hornik, K., Ihaka, R., McWhite, C. D., Murrell, P., Stauffer, R. & Wilke, C. O. (2020), ‘colorspace: A Toolbox for Manipulating and Assessing Colors and Palettes’, *Journal of Statistical Software* **96**(1), 1–49. <https://www.jstatsoft.org/index.php/jss/article/view/v096i01>.

Zhu, H. (2024), *kableExtra: Construct Complex Table with kable and Pipe Syntax*. R package version 1.4.0. <https://CRAN.R-project.org/package=kableExtra>.